

Data Visualization & Analytics Assignment

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Course: Data Visualization

1. Introduction

This report analyzes Pakistan's Largest E-Commerce Dataset to understand sales trends, customer behavior, and operational performance. The dataset includes order details such as payment methods, product categories, and order statuses.

2. Data Understanding

The dataset contains 584509 rows and 17 columns. It includes transaction data from 2016 to 2018. Key columns include 'status', 'created_at', 'price', and 'category_name_1'.

3. Data Cleaning Summary

Cleaning steps performed:

- Converted 'created_at' to datetime format.
- Removed rows with missing statuses or dates.
- Filled missing categories with 'Unknown'.
- Dropped irrelevant columns like 'sales_commission_code'.
- Removed invalid orders (negative prices or zero quantity).

4. Visualizations and Insights

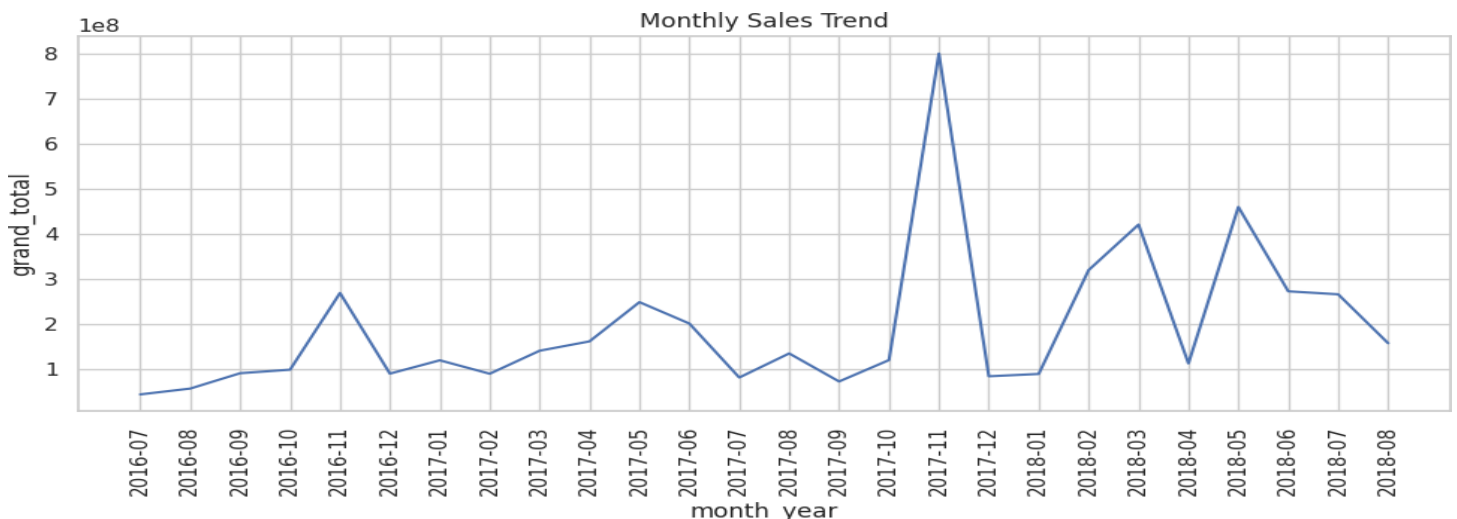


Fig 1: Monthly sales show significant spikes in November (Black Friday sales).

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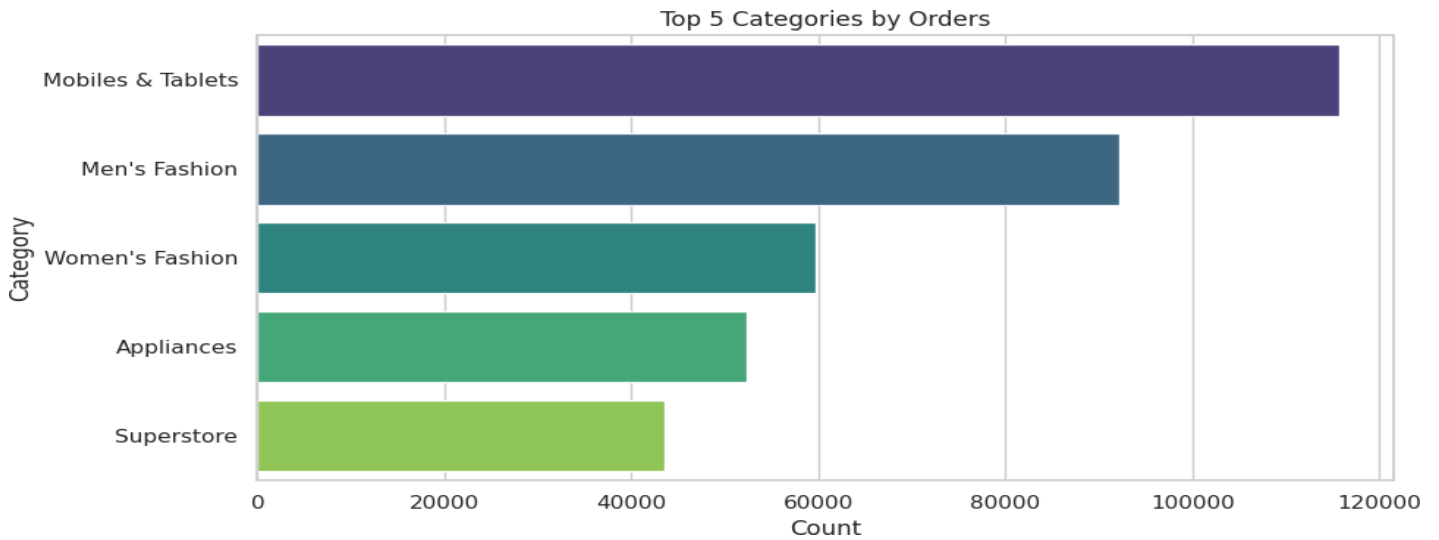


Fig 2: Mobiles & Tablets and Men's Fashion are the most ordered categories.

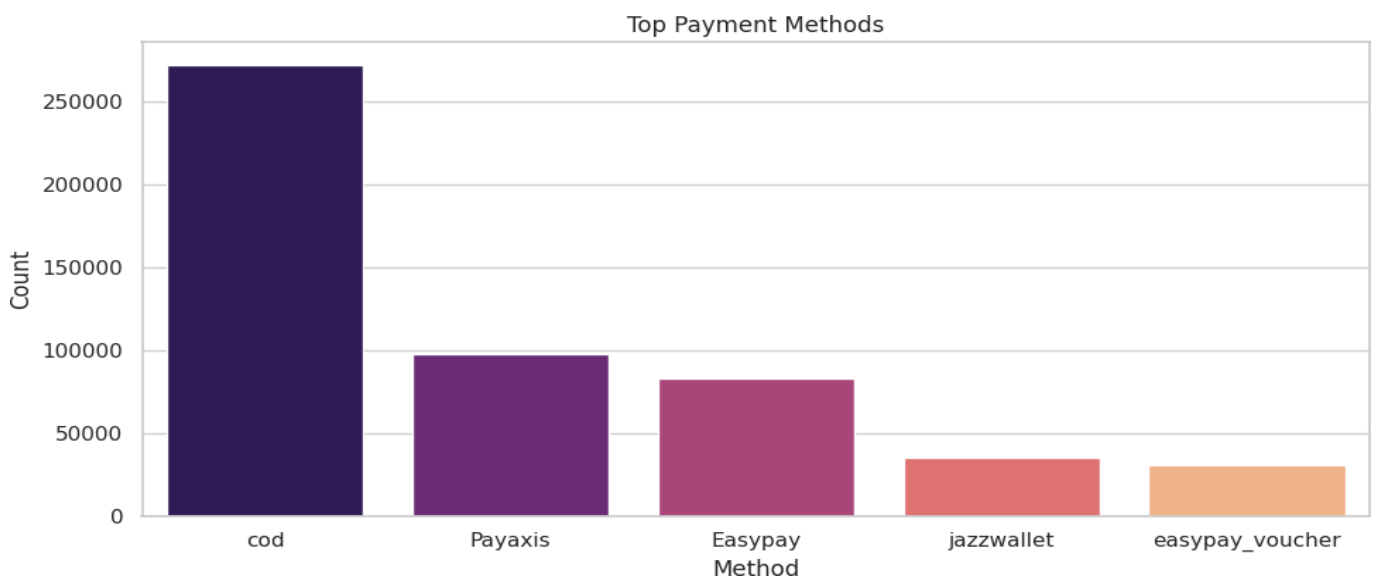


Fig 3: Cash on Delivery (COD) is the dominant payment method.

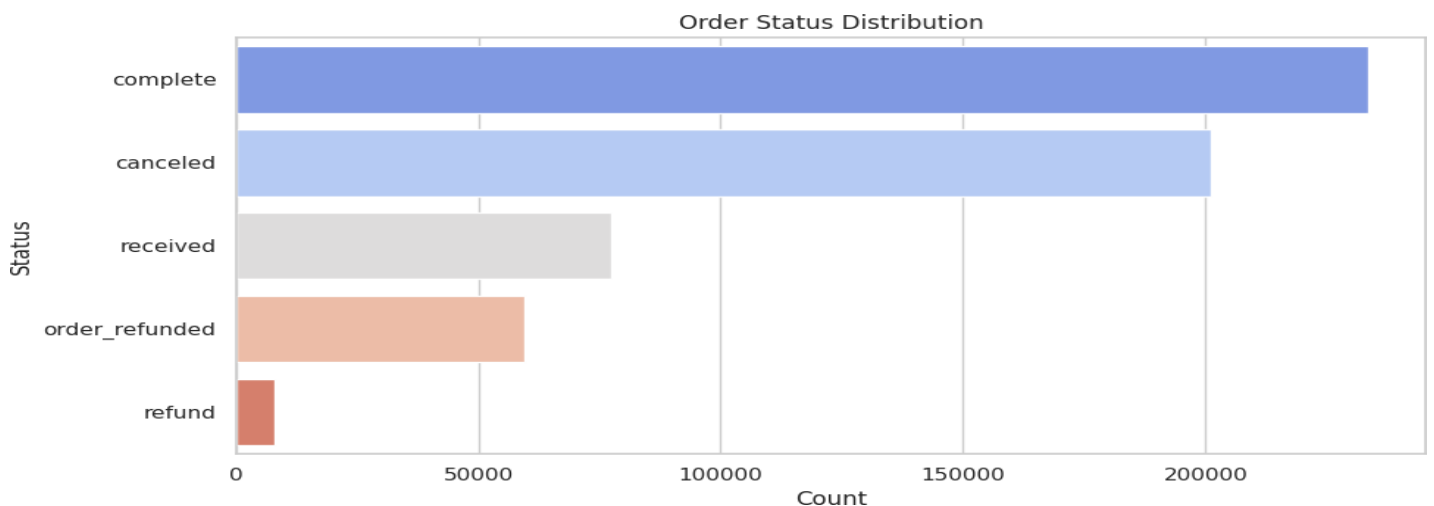


Fig 4: A high number of orders are canceled or refunded.

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5. Key Statistical Findings

- Total Revenue: 4,986,291,235.57 PKR
- Total Orders: 408771
- Average Order Value: 12,198.25 PKR
- Cancellation Rate: 45.99%

6. Conclusion

The analysis reveals that while the e-commerce platform generates significant revenue, it faces challenges with high cancellation rates (~44%). The reliance on Cash on Delivery remains a major factor. The business sees strong seasonality, particularly in late-year sales.